# Amina Saif

M.Sc. Cybersecurity Student (BTU Cottbus-Senftenberg) | Network Security & Linux

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## Profile

Master's student in Cybersecurity with 3 years of professional experience across OT/industrial network security and backend software development. Hands-on with network programming (POSIX/UNIX sockets, MQTT, WebSockets), traffic analysis (Wireshark), Linux, and Python/Ruby/Java scripting. Bachelor's thesis on network analysis, penetration testing and vulnerability exploitation.

## Education

**M.Sc. Cybersecurity** — Oct 2025 – present
*Brandenburgische Technische Universität Cottbus-Senftenberg* — Cottbus, Germany

**B.Sc. Information Technology** — Oct 2017 – Aug 2021
*University of the Punjab, Information Technology College (PUCIT)* — Lahore, Pakistan

- Final grade: 3.55/4.00
- Thesis: *Network Analysis, Penetration Testing and Vulnerability Exploitation*

## Skills

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|---|---|
| **Programming & scripting** | Python, Ruby, Java, Scala, JavaScript, SQL (MySQL, MSSQL, PL/SQL) |
| **Network programming** | POSIX/UNIX sockets, MQTT, WebSockets, REST and SOAP APIs, IIoT communication |
| **Networks & protocols** | TCP/IP, network segmentation, CCNA-level routing and switching, Wireshark |
| **Linux/UNIX** | Linux command line, VMware ESXi 7, Docker, Git/GitHub/GitLab, Jenkins |
| **Security tools & practice** | Metasploit, vulnerability mitigation, IDS/IPS, threat detection, OS hardening, Active Directory, Claroty CTD/EMC |
| **Languages** | Urdu (native), English (C1), German (A1) |

## Professional Experience

**Software Engineer** — Dec 2023 – Sep 2025
*PakWheels* — Lahore, Pakistan

- **API & Database Optimization:** Optimized database queries, indexing strategies and backend execution paths across high-traffic services, delivering response-time improvements of up to approximately 2× across targeted workflows.
- **MySQL Modernization:** Contributed to migrating the primary database from MySQL 5.7 to MySQL 8, contributing approximately $3K/month in infrastructure and operational savings.
- **Infrastructure Optimization:** Implemented infrastructure improvements that contributed approximately $1K/month in additional savings.
- **Production Reliability:** Resolved critical issues across ad posting, featuring and mobile verification workflows, contributing approximately $1.3K/month in operational savings.
- **Caching & Background Processing:** Implemented in-memory and query-result caching alongside background processing, improving efficiency and response times by approximately 40–70% for frequently accessed workflows.
- **Rails Modernization:** Upgraded the primary Ruby on Rails application from Rails 3.2 to Rails 4.2, improving maintainability and stability of the legacy backend.
- **Engineering Leadership:** Managed 2 interns and 1 junior software engineers while participating in peer code reviews and supporting backend quality standards.

**Cybersecurity Engineer (OT Security)** — Sep 2022 – Dec 2023
*Intech Process Automation* — Lahore, Pakistan

- Planned security basline and a justified network design for Industrial sites before protection measures could be planned.
- Built hardware and software inventories from documentation review, performed gap assessments against target frameworks and standards, carried out risk assessments to establish the rationale for network segmentation, developed cybersecurity baselines for existing OEM systems, documented all projects.
- Gave clients a structured view of their security posture and a risk-based basis for segmentation and hardening decisions.

## Projects

**UEP Deployment & Implementation Project** Jun 2023 – Aug 2023

- Deploy and harden the core security infrastructure and assess suspicious software.
- Deployed a Domain Controller, WSUS server, antivirus server and rugged firewalls, hardened operating systems and applications, assessed suspicious software.
- Delivered a hardened, centrally managed environment with controlled patching, endpoint protection and perimeter filtering.

**Claroty IDS Support (KPO)** Mar 2023 – Sep 2023

- Investigate the problems and find the underlying cause related to installed Claroty SRA, EMC and IDS components in Oil and Gas Plant site.
- Analysed the installed components, worked with the Claroty team to isolate the root cause, and devised a mitigation plan.
- Identified the root of the recurring faults and provided a plan to resolve them, restoring reliable network monitoring.

**Bachelor Thesis: Network Analysis, Penetration Testing & Vulnerability Exploitation** 2021

- Study how network analysis supports penetration testing and vulnerability exploitation.
- Analysed network traffic and used penetration-testing tooling (e.g., Wireshark, Metasploit) to identify and exploit vulnerabilities in a test setting.
- Completed the thesis with a documented methodology; B.Sc. final grade 3.55/4.00.

**Web Forms Data-Analytics Application, Petroleum Development Oman** Oct 2022 – Dec 2023

- Build a reliable C# ASP.NET Web Forms application backed by an efficient database for Tableau integration and effective analysis.
- Gathered requirements with stakeholders, designed database structures for fast storage and retrieval, built the interface, tested and debugged, and reported progress regularly.
- Delivered a stable application that supported the client's analytics needs.

## Certifications

- Cisco Certified Network Associate (CCNA), Sep 2023
- Claroty Cybersecurity Analyst (CCA 601) and Claroty Implementation Engineer (CIE 401), 2023
- PTC ThingWorx Associate Certification, 2023
- Udemy: Backend Performance Optimization, Mar 2025

## Additional

- EUNICE Summer School on Cybersecurity, June 2026, Poznan University of Technology, Poland.
- Xperience Workshop for IIoT, Dubai: automation, analytics and IIoT trends.
- Availability: flexible working hours, remote work possible.